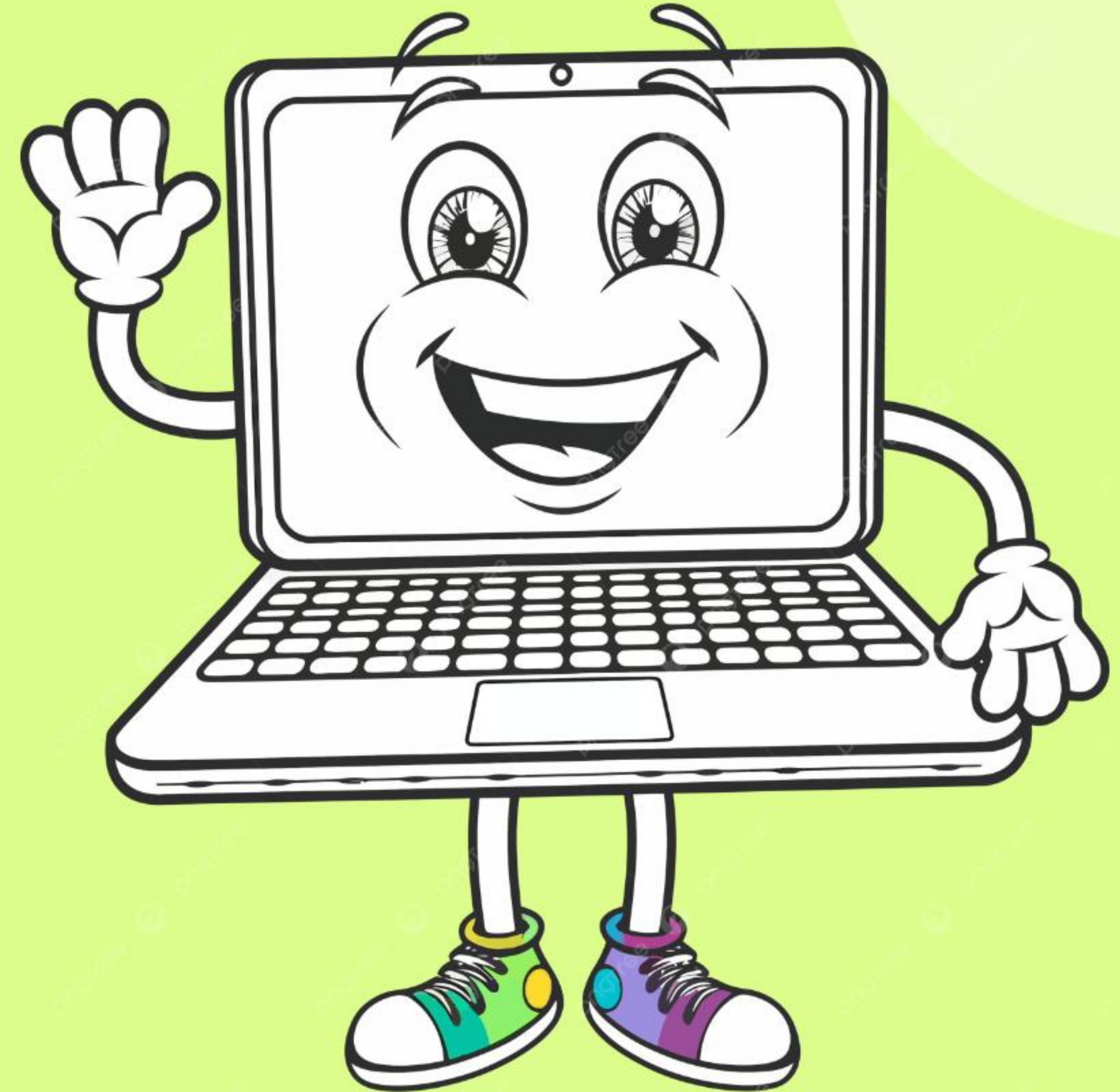


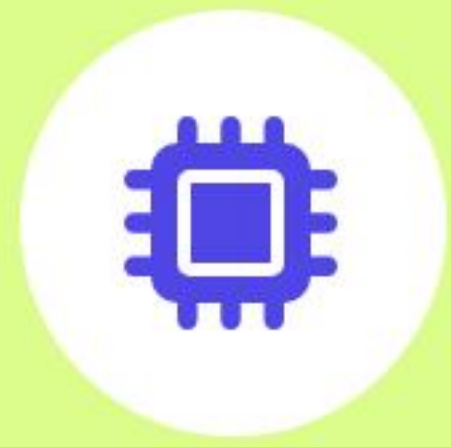
Computer Science

Hardware Heroes

Grade 4 Discovery Lab



Hardware vs. Software



Hardware

The physical parts of the computer that you can **see and touch**.

Example: Keyboard, Monitor, CPU.



Software

Instructions that tell the computer what to do. You **cannot touch** it!

Example: Games, MS Paint, Windows.

Input: Giving Instructions



Keyboard

Enter data using keys like letters, numbers, and symbols.



Mouse

A pointing device used to select objects and draw.



Microphone

Records your voice or sounds into the computer system.

Output: Seeing Results



Monitor

Displays visual information like movies and text.



Printer

Gives a "Hard Copy" of your work on paper.



Speakers

Lets you hear sounds, music, and voice from your PC.

Workout: Which device should I use?

1. Movie on a large screen?

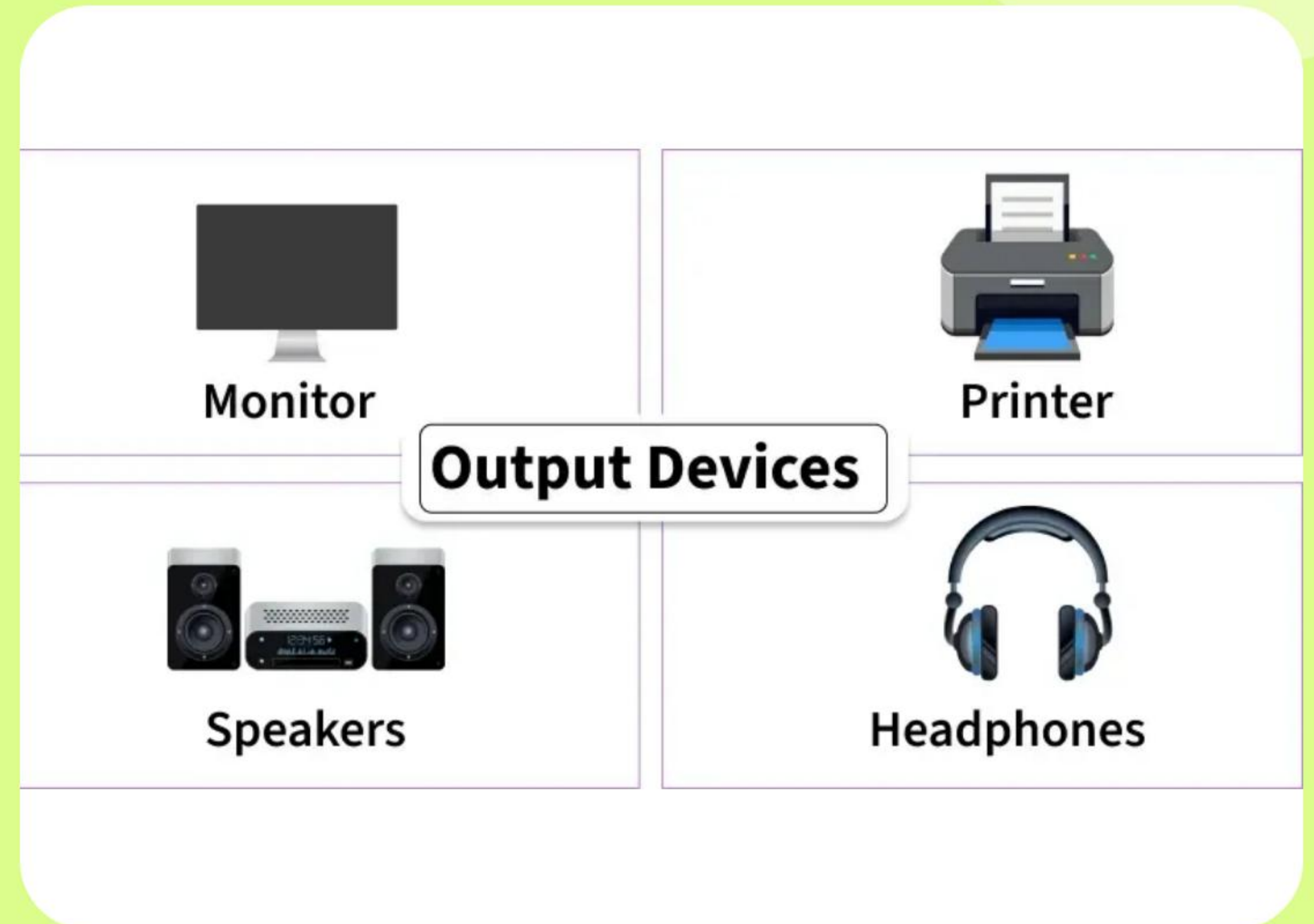
→ Use a **PROJECTOR**

2. Print lots of text in high quality?

→ Use a **LASER PRINTER**

3. Listen to music without disturbing anyone?

→ Use **HEADPHONES**



What is Computer Memory?

Just like you remember your birthday, a computer has **Memory** to store information.

It can store things **Temporarily** (short time) or **Permanently** (long time).

Two Types:

- Internal Memory (Inside)
- External Memory (Outside)

 Memory Chips



Internal Memory: RAM and ROM



RAM

Random Access Memory

Stores data temporarily while you work. It is erased when the computer is turned off!



ROM

Read Only Memory

Stores data permanently. It holds start-up instructions and is NOT erased when you turn it off.

External Memory

This is also called **Secondary Memory**.

We use it to store information for a long time. It is not erased even when the power goes off!

These are called **Storage Devices**.



Examples:

- Hard Disk (Inside CPU box)
- Pen Drive
- CD / DVD
- Memory Card

Super Storage: Memory Card



The Tiny Hero

A **Memory Card** is very small but can store many photos and videos.

Commonly used in:

- Digital Cameras
- Smartphones
- Tablets

It is portable, so you can take your data anywhere!

Quick Look: Storage Devices

Device	Type	Where is it?
Hard Disk	External/Secondary	Inside the CPU box
Pen Drive	External/Secondary	Plugged into USB port
CD/DVD	External/Secondary	Inserted in Disk Drive
RAM	Internal/Primary	On the Motherboard

Quiz Time! (1-3)

Q1

Which device is used to watch movies on a big screen?

Check Answer

Q2

Is a Mouse Hardware or Software?

Check Answer

Q3

Which internal memory gets erased when power goes off?

Check Answer

Quiz Time! (4-6)

Q4

What is also known as
"Secondary Memory"?

Check Answer

Q5

What device gives you a "Hard
Copy"?

Check Answer

Q6

Is ROM erased when you turn off
the computer?

Check Answer

Quiz Time! (7-9)

Q7

What tiny storage is used in digital cameras?

Check Answer

Q8

Which printer gives high-quality text prints?

Check Answer

Q9

Which hardware lets you listen to music alone?

Check Answer

Great Job! Keep Learning!

Explore the technology around you every day! 🚀



Image Sources



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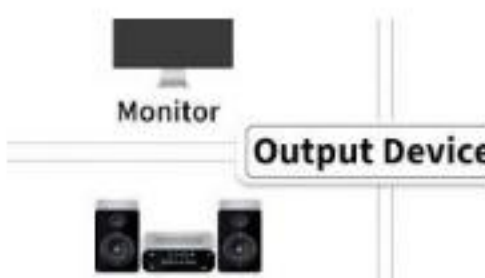
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